

Senior Citizen Identification System

1) Project Overview

This project implements a Multi-Person Age-Gender Detection and Senior Citizen Identification System for video and real-time webcam feeds.

The system combines Deep Learning and Computer Vision techniques to:

1. Detect multiple people in a video or real-time webcam feed.
2. Predict each detected person's age using a CNN.
3. Predict each detected person's gender using a CNN.
4. Track each person across frames so a single visit is only logged once.
5. Flag anyone with predicted age above 60 as a Senior Citizen.
6. Record age, gender, and time of visit in a CSV/Excel file.
7. Provide a graphical user interface (GUI) for live webcam and uploaded-video use.

2) Problem Statement

Standard face-analysis systems predict age and gender for a single, already-cropped photo. This task extends that baseline behaviour to full video/webcam streams containing multiple, moving people, and adds an automated visit-logging requirement on top.

The objective of this project is to develop a machine learning system capable of:

1. Detecting every person's face in a video or webcam frame, not just one.
2. Predicting age and gender for each detected face using CNNs trained from scratch.
3. Marking anyone whose predicted age exceeds 60 as a senior citizen.
4. Tracking each person across frames so they are logged once per visit, not once per frame.
5. Storing age, gender, and time of visit in a CSV/Excel file.

3) Dataset

Age, Gender and Ethnicity (Face Data) CSV

Dataset Source:

Kaggle - [nipunarora8/age-gender-and-ethnicity-face-data-csv](#)

Age and gender labels, along with the face pixel data itself, are already provided as columns in the CSV, so no filename parsing or separate annotation step is required:

age, gender, pixels

Here, gender 0 = Male and 1 = Female.

Dataset Structure:

- 23,705 images available in the full dataset, of which 2,395 (about 10%) are seniors (age > 60)
- 15,000 images used (a subset of the ~23,705 available) to keep training fast
- 12,750 training images / 2,250 testing images (85/15 split)

4) Methodology

1. Data Preprocessing

- Face pixel strings decoded back into 48x48 grayscale images.
- Pixel values normalized to [0,1].
- Age and gender labels read directly from the CSV's own columns.
- Dataset split into training and testing sets (85/15).

2. Shared Age-Gender CNN

A single Convolutional Neural Network (CNN) was trained entirely from scratch (no pretrained weights), with two output heads sharing the same convolutional backbone - one for gender, one for age.

Model Architecture:

- Convolution Layer (32 filters)
- Max Pooling Layer
- Convolution Layer (64 filters)
- Max Pooling Layer
- Convolution Layer (128 filters)
- Max Pooling Layer
- Dense Layer (128 units)
- Dropout Layer (0.3)
- gender_output: Dense Layer (1 unit, Sigmoid)
- age_output: Dense Layer (1 unit, Linear)

3. Multi-Person Face Detection

OpenCV's Haar cascade face detector locates every face in a video frame, tuned with minSize=(80, 80) and minNeighbors=10 so small background clutter and patterned clothing aren't mistaken for faces. Any detected region that is visually too flat to plausibly be a face (a blank wall, a glare spot) is discarded before it reaches the tracker.

4. Tracking and Visit Logging

A lightweight tracker matches faces frame-to-frame by how much their bounding boxes overlap (IOU), so the same visitor isn't logged repeatedly, and survives roughly 1.3 seconds of a person being blocked from view.

For each tracked person:

1. Predict age and gender using the CNN, once per frame.
2. Average the predictions over several consecutive frames for a stable reading.
3. If the averaged predicted age is above 60:
 - o Mark the person as Senior Citizen
4. Else:
 - o Log the visit as a regular (non-senior) visitor

Once a track is confirmed, the visit is appended to visit_log.csv and mirrored into visit_log.xlsx.

5) GUI Development

Gradio was used to develop a graphical user interface with two tabs.

Features:

- Live Webcam tab - streams directly from the viewer's own browser camera.
- Upload Video tab - processes an uploaded video file end-to-end.
- Annotated detections with bounding boxes and senior-citizen labels.
- Downloadable visit_log.csv and visit_log.xlsx after each run.

6) Results

Age-Gender Model

Gender Test Accuracy: 85.2%

Gender F1-score: 0.847

Age Test MAE: 7.44 years

Age R2: 0.729

Senior-Citizen Flag (predicted age > 60 vs. true age > 60)

Accuracy: 93.7%

Precision: 0.739

Recall: 0.526

F1-score: 0.614

Multi-Person Detection Verification

Successfully verified on:

- Real, unlabeled stock video footage of a mall walkway containing multiple people, including one visibly older individual.
- Detection and tracking parameters tuned after observing the pipeline both over-count and under-count visitors on the same test clip.

Example: on a 6.7-second test video, tuning the face detector's minimum size and neighbor-confidence settings reduced the logged visit count from 37 down to approximately 8, while correctly catching the senior citizen present in the footage for the first time.

7) Visualizations

Included in notebook/repository:

- Age Distribution Histogram
- Gender Distribution Bar Chart
- Age-Gender Model Training Accuracy Graph
- Age-Gender Model Training Loss Graph
- Gender Model Confusion Matrix
- Gradio input video

8) Technologies Used

- Python
- TensorFlow / Keras
- OpenCV
- NumPy
- Pandas
- Matplotlib
- Scikit-Learn
- Gradio

9) Future Improvements

- Train on the full dataset instead of a 15,000-image subset.
- Use data augmentation.
- Use transfer learning to reduce age-estimation MAE.
- Replace the Haar cascade with a modern deep-learning face detector for crowded or poorly lit scenes.
- Add motion prediction to the tracker for more robust occlusion handling.
- Add confidence scores alongside each logged prediction.
- Add a dashboard summarizing visitor statistics over time.
- Deploy the application using Hugging Face Spaces, Streamlit, or Flask.

10) Conclusion

This project successfully develops a CNN-based Senior Citizen Identification System that detects multiple people in a video or webcam feed, predicts their age and gender, flags anyone above 60 as a senior citizen, and records their age, gender, and visit time. The age-gender CNN reached 85.2% gender accuracy and a 7.44-year age MAE on a 15,000-image subset of the Age, Gender and Ethnicity (Face Data) CSV dataset, with the senior-citizen flag itself reaching 93.7% accuracy. The multi-person detection, tracking, and logging pipeline was verified against real test video footage, with detection and tracking parameters deliberately tuned to reduce both over-counting and under-counting of visitors, confirming the system works as specified.